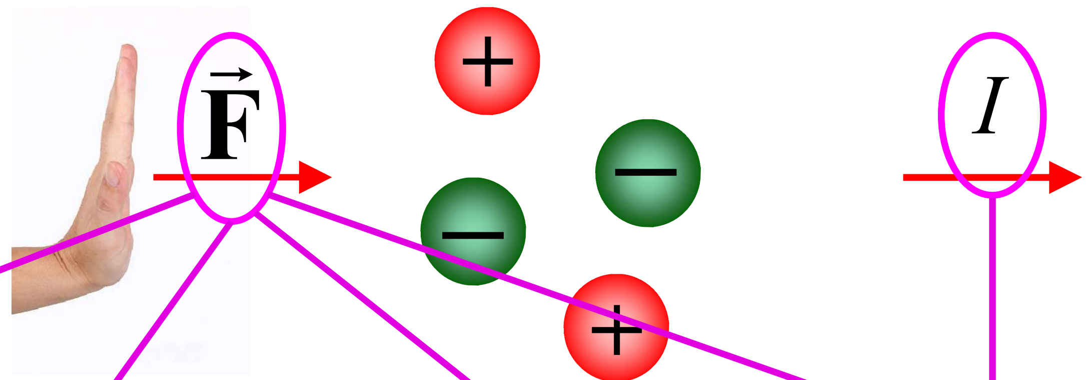
Moving charges: Current



Mechanical
(*e.g., motion of charged particles in a cloud due to wind!*)

Electromagnetic
(*due to an electromagnetic field*)

Chemical
(*e.g., by the battery*)

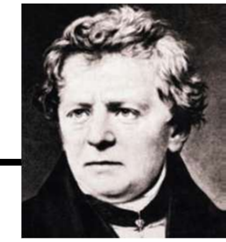
Thermal
(*e.g., thermionic emission of electrons from hot surfaces*)

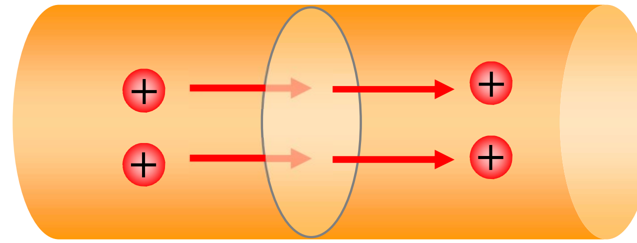
Conduction current
(*currents in conductors*)

Convection current

⋮

Ohm's Law





$\vec{\mathbf{J}} \propto \vec{\mathbf{f}}$ (force per unit charge)

$\vec{\mathbf{J}} = \sigma \vec{\mathbf{f}}$ σ : conductivity

$\dfrac{1}{\sigma} = \rho$ Resistivity (Ω.m)

usually negligible, except in plasma

generally taken as ∞

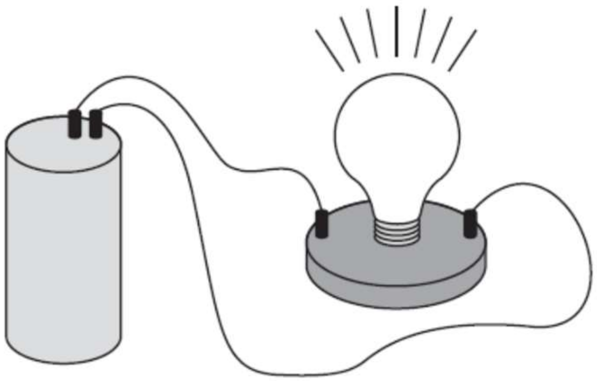
Conductivity of the metals $= 10^{23} \times$ Conductivity of insulators

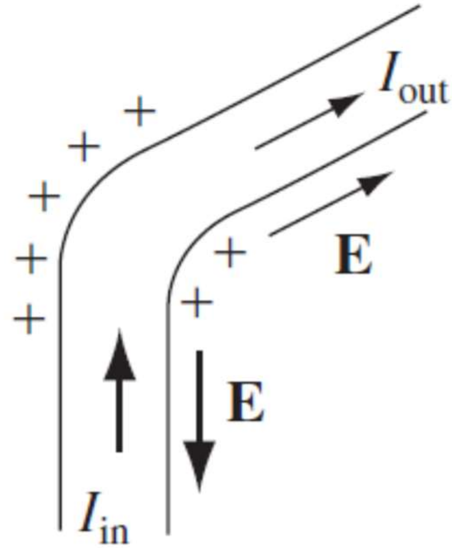
$\vec{\mathbf{J}} = \sigma\,(\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}})$

$\boxed{\vec{\mathbf{J}} = \sigma\,\vec{\mathbf{E}}}$ **Ohm's Law**

Material	Resistivity (Ω.m)
Copper	$\sim 10^{-8}$
Hard Rubber	$\sim 10^{15}$

Electromotive Force





$$I_{in} > I_{out}$$

$$I_{in} = I_{out}$$

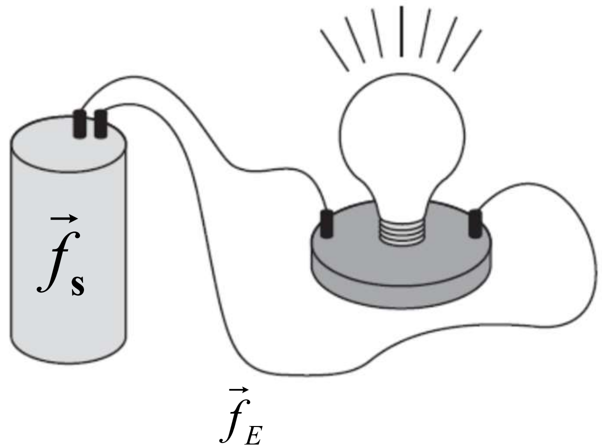
Equilibrium is established…

System automatically self corrects itself to keep current uniform.

Basic questions:

- Current is same all the way along the loop at any given moment. Why?
- Driving agent is inside the battery. Who is doing the pushing in the rest of the circuit?
- How does it happen that this push is exactly correct to produce same current in each segment?

Electromotive Force



Driving forces:

– by source $\vec{f}_s$

– by electrostatic field $\vec{\mathbf{E}}$

Communicates the influence of the source at a distant point in the circuit

The force per unit charge driving the current is $\vec{f}_{net} = \vec{f}_s + \vec{\mathbf{E}}$

$$\oint \vec{f}_{net} \cdot d\vec{\mathbf{l}} = \oint \vec{f}_s \cdot d\vec{\mathbf{l}} + \cancel{\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}}}$$

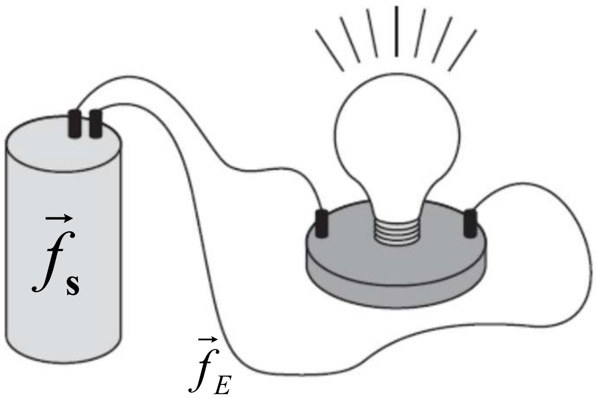
$$\Rightarrow \oint \vec{f}_{net} \cdot d\vec{\mathbf{l}} = \oint \vec{f}_s \cdot d\vec{\mathbf{l}} = \mathcal{E}$$

For ideal source (e.g., resistance-less battery) $\sigma \rightarrow \infty$

$$\vec{\mathbf{J}} = \sigma \vec{\mathbf{f}} \quad \Rightarrow \quad \vec{f}_{net} = 0 \quad \Rightarrow \quad \vec{f}_s = -\vec{\mathbf{E}}$$

P.D. between two terminals $V = -\int_b^a \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = \int_b^a \vec{f}_s \cdot d\vec{\mathbf{l}} = \boxed{\mathcal{E} = \oint \vec{f}_s \cdot d\vec{\mathbf{l}}}$

Electromotive Force



Driving forces:

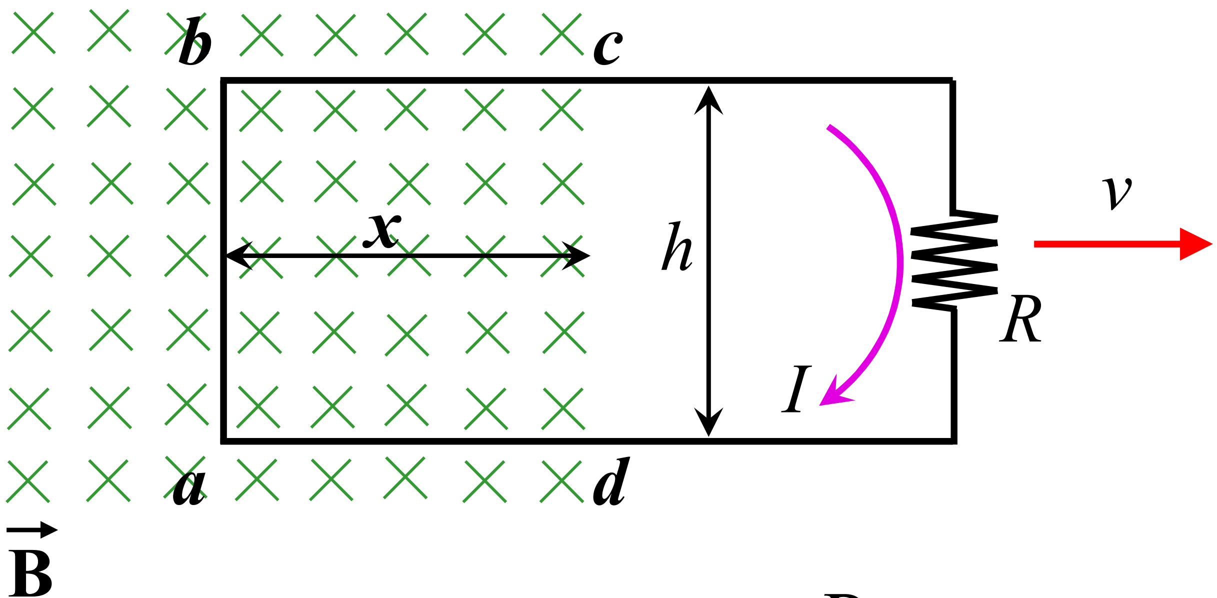
– by source $\vec{f}_s$

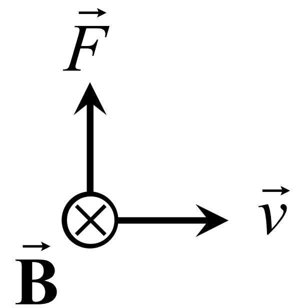
– by electrostatic field $\vec{\mathbf{E}}$

$$V = -\int_b^a \vec{\mathbf{E}} \cdot \mathbf{d}\vec{\mathbf{l}} = \int_b^a \vec{f}_s \cdot \mathbf{d}\vec{\mathbf{l}} = \boxed{\mathcal{E} = \oint \vec{f}_s \cdot \mathbf{d}\vec{\mathbf{l}}}$$

- $\mathcal{E}$: work done per unit charge by the source
- Source of EMF establishes and maintains the voltage difference equal to the EMF
- Resulting electrostatic field drives current around the rest of the circuit

Motional EMF: Electric Generator





$$\left|\vec{f}_{mag}\right| = \frac{qvB}{q} \quad \Rightarrow \quad \mathcal{E} = \oint \vec{f}_{mag} \cdot d\vec{l} = vBh$$

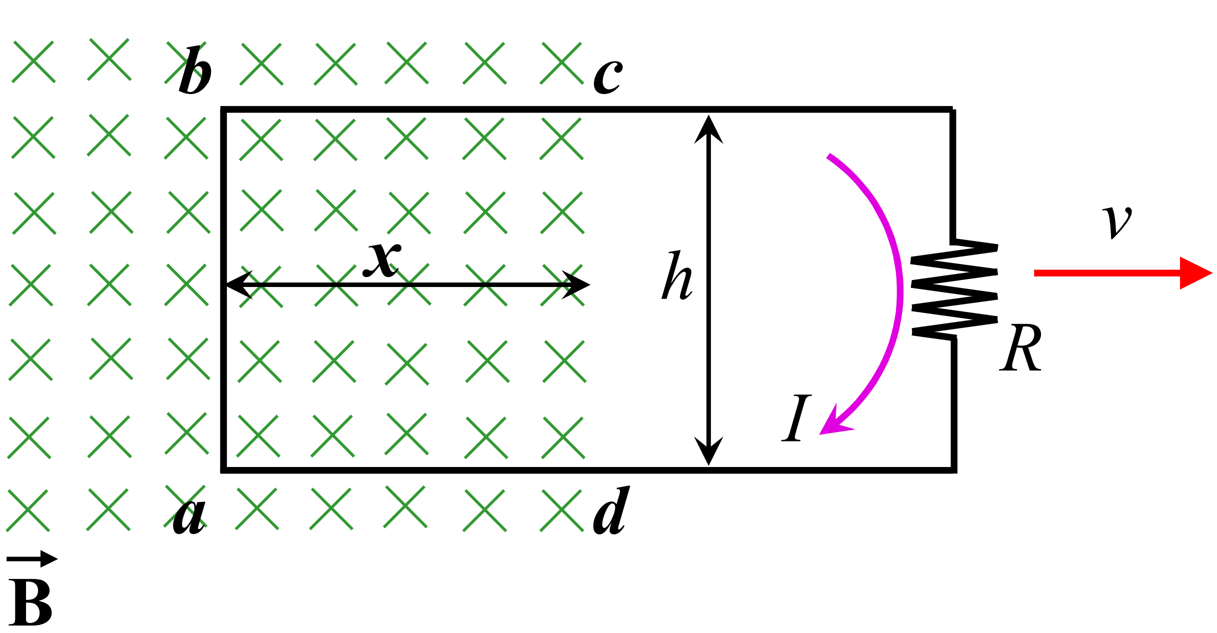
Puzzle

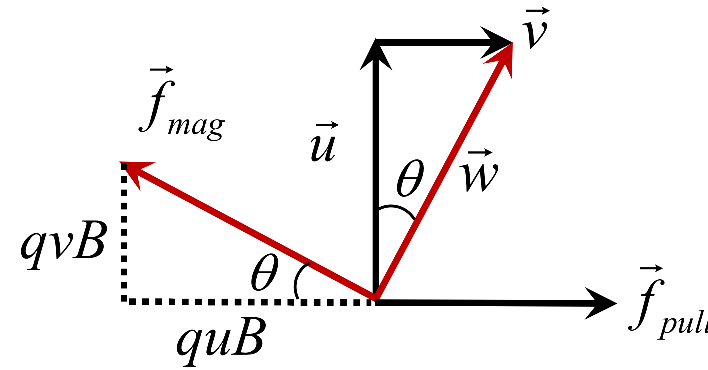
Magnetic force is responsible for establishing EMF. But is not doing any work for sure!

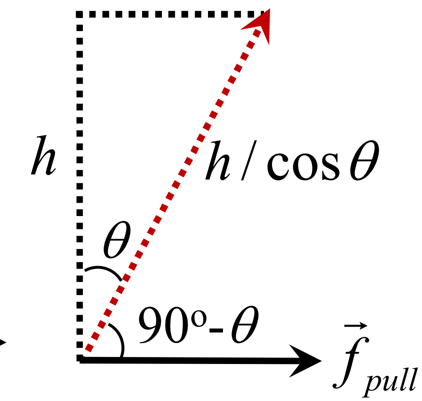
Then, who is supplying the energy dissipated into *R*?

Person who is pulling the loop!

Motional EMF: Electric Generator







Work done per unit charge by external agent $= \int \vec{f}_{pull} \cdot \mathbf{d}\vec{\mathbf{l}}$

$$= uB\frac{h}{\cos\theta}\cos(90^\circ - \theta)$$

$$= uBh\tan\theta$$

$$= \cancel{u}Bh\left(\frac{v}{\cancel{u}}\right) = Bhv = \mathcal{E}!$$

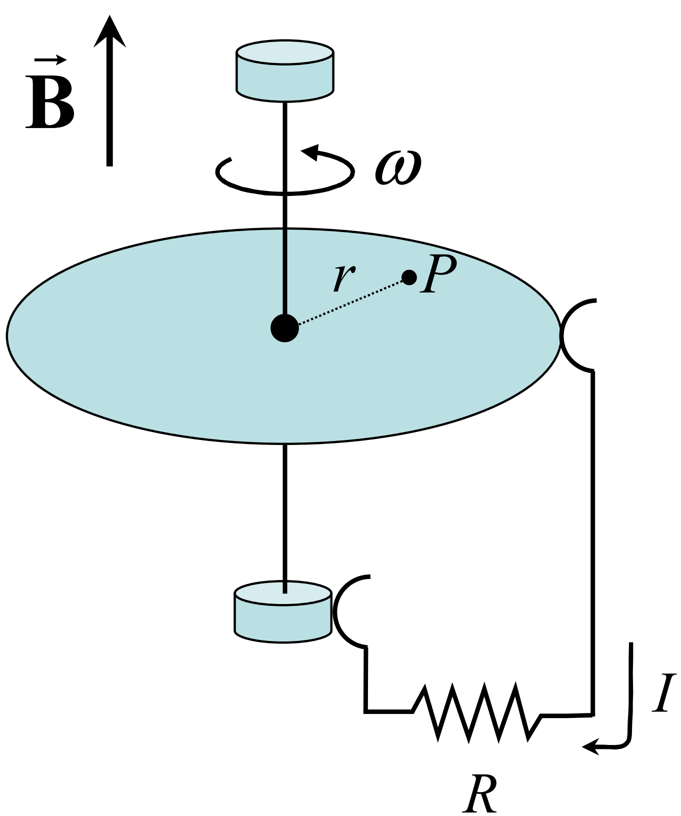
$$\Phi = \int \vec{\mathbf{B}} \cdot \mathbf{d}\vec{\mathbf{a}} = Bhx$$

$$\frac{d\Phi}{dt} = Bh\frac{dx}{dt} = -Bhv$$

$$\boxed{\mathcal{E} = -\frac{d\Phi}{dt}}$$

Motional EMF: Example

A metal disk of radius a rotates with angular velocity ω about a vertical axis, through a uniform field **B**, pointing up. A circuit is made by connecting one end of a resistor to the axle and the other end to a sliding contact, which touches the outer edge of the disk. Find the current in the resistor.



Speed of point P: $v = r\omega$

Force per unit charge: $\vec{f}_{mag} = \vec{\mathbf{v}} \times \vec{\mathbf{B}}$

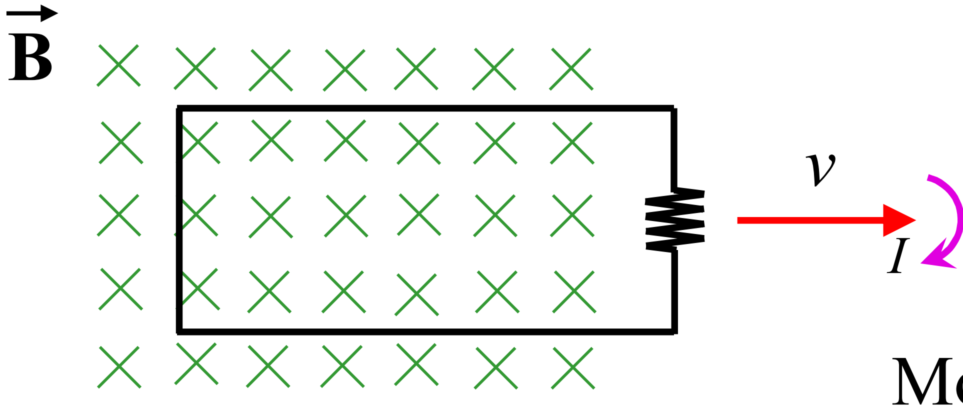
$$= r\omega B\,\hat{\phi} \times \hat{z}$$

$$= r\omega B\,\hat{r}$$

$$\mathcal{E} = \oint \vec{f}_{mag} \cdot \mathbf{d}\vec{\mathbf{l}} = \int_0^a r\omega B\hat{r} \cdot d\vec{r} = \frac{\omega B a^2}{2}$$

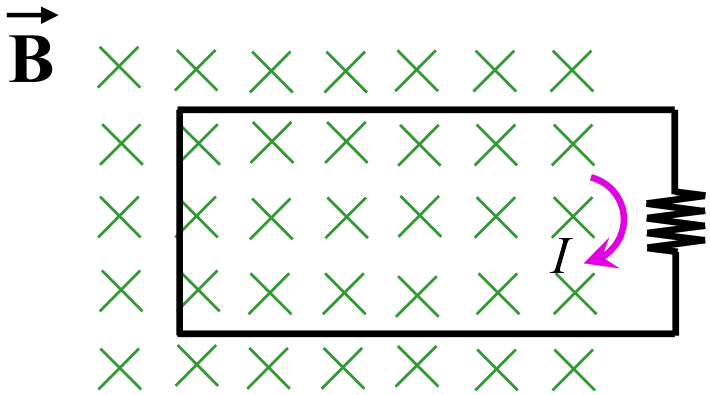
$$\Rightarrow \quad I = \frac{\mathcal{E}}{R} = \frac{\omega B a^2}{2R}$$

Faraday's Law of EM Induction (1831)



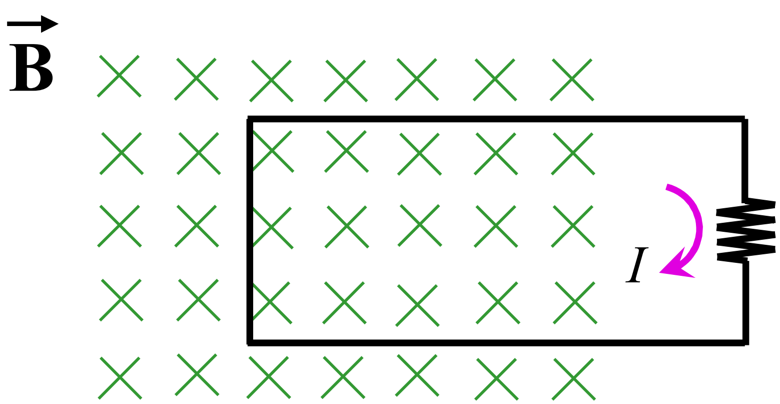
Motional EMF $\mathcal{E} = -\dfrac{d\Phi}{dt}$

Magnetic force sets up EMF



$\cancel{\vec{f}_{mag} = \vec{\mathbf{v}} \times \vec{\mathbf{B}}}$

Changing magnetic field induces an electric field



Magnetic flux through the loop changes in all the three exps $\longrightarrow$ Induced EMF

Faraday's Law of EM Induction

$$\boxed{\mathcal{E} = \oint \vec{\mathbf{E}} \cdot \mathbf{d}\vec{\mathbf{l}} = -\frac{d\Phi}{dt}}$$

$$= -\frac{d}{dt}\int \vec{\mathbf{B}} \cdot \mathbf{d}\vec{\mathbf{a}}$$

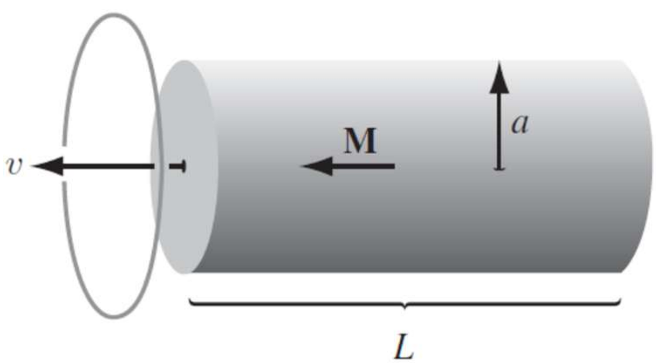
$$\Rightarrow \quad \int (\vec{\nabla} \times \vec{\mathbf{E}}) \cdot \mathbf{d}\vec{\mathbf{a}} = -\int \frac{d\vec{\mathbf{B}}}{dt} \cdot \mathbf{d}\vec{\mathbf{a}}$$

$$\Rightarrow \quad \boxed{\vec{\nabla} \times \vec{\mathbf{E}} = -\frac{d\vec{\mathbf{B}}}{dt}}$$

Induced current and associated magnetic flux have directions such that they oppose change of the flux. $\longrightarrow$ Lenz's rule

Faraday's Law of EM Induction

Example 1: A long cylindrical magnet of length L and radius a carries a uniform magnetization **M** parallel to its axis. It passes at constant velocity v through a circular wire ring of slightly larger diameter. Graph the emf induced in the ring, as a function of time.



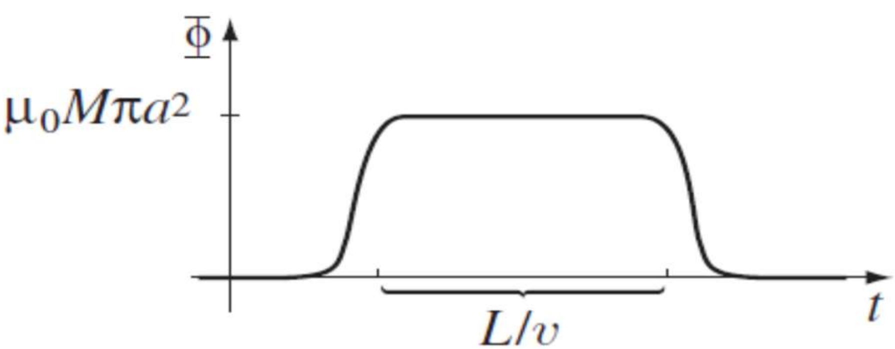
$$\vec{\mathbf{J}}_b = \vec{\nabla} \times \vec{\mathbf{M}} = \mathbf{0}$$

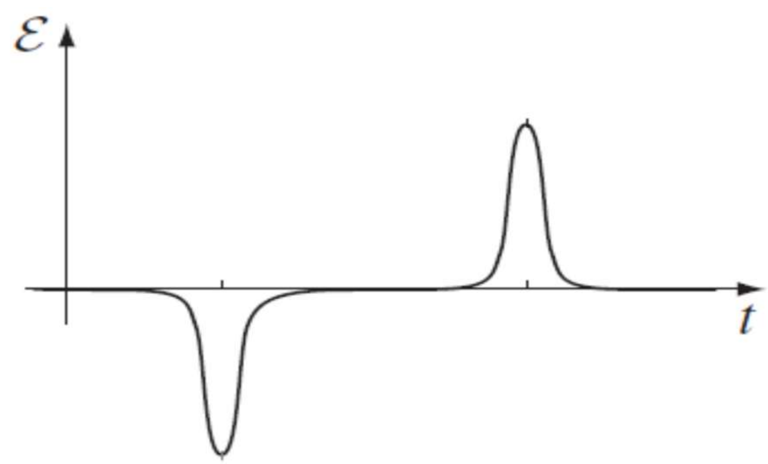
$$\vec{\mathbf{K}}_b = \vec{\mathbf{M}} \times \hat{n} = M\ \hat{z} \times \hat{r} \ = M\ \hat{\phi}$$

Long solenoid, with field along z

$$\vec{\mathbf{B}} = \mu_0 n I\ \hat{z} = \mu_0 K_b\ \hat{z}\ = \mu_0 M\ \hat{z}$$

$$\Phi = \mu_0 M\ \pi\ a^2$$





HW: Example 7.6

Summery so far…

There are two distinct kinds of electric fields:

– associated with electric charge

$$\vec{\nabla}\cdot\vec{\mathbf{E}}=\frac{\rho}{\varepsilon_0}\qquad \vec{\nabla}\times\vec{\mathbf{E}}=0$$

– associated with changing magnetic field

$$\vec{\nabla}\cdot\vec{\mathbf{E}}=0\qquad \vec{\nabla}\times\vec{\mathbf{E}}=-\frac{\partial\vec{\mathbf{B}}}{\partial t}$$

Additionally, magnetic fields are obtained through:

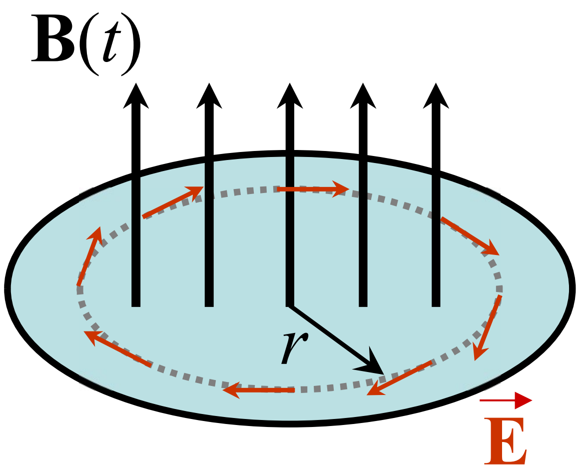
$$\vec{\nabla}\cdot\vec{\mathbf{B}}=0\qquad \vec{\nabla}\times\vec{\mathbf{B}}=\mu_0\vec{\mathbf{J}}$$

or

$$\oint\vec{\mathbf{E}}\cdot\mathbf{d}\vec{\mathbf{l}}=-\frac{d\Phi}{dt}\qquad \oint\vec{\mathbf{B}}\cdot\mathbf{d}\vec{\mathbf{l}}=\mu_0 I_{en}$$

Faraday's Law of EM Induction

Example 2: A uniform magnetic field **B**(t), pointing straight up, fills the shaded circular region as shown in the following figure. If **B** is changing with time, what is the induced electric field?



$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = -\frac{d\Phi}{dt}$$

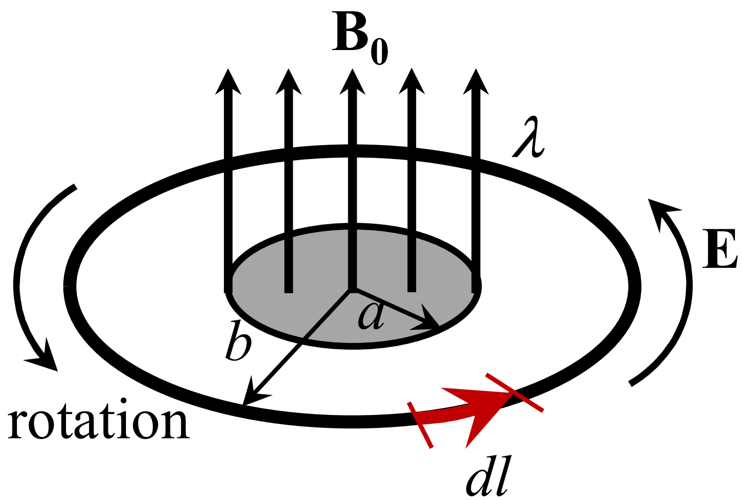
$$E(2\pi r) = -\frac{d}{dt}[\pi r^2 B(t)]$$

$$= -\pi r^2 \frac{dB}{dt}$$

$$\Rightarrow \quad \vec{\mathbf{E}} = -\frac{r}{2}\frac{dB}{dt}\hat{\phi}$$

Faraday's Law of EM Induction

Example 3: A line charge λ is glued onto the rim of a wheel of radius b, which is then suspended horizontally. There is a uniform magnetic field $\mathbf{B}_0$ in the central region up to radius a. Now field is turned off. What happens then afterwards?



$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = -\frac{d\Phi}{dt} \Rightarrow \oint E\, dl = -\pi a^2 \frac{dB}{dt}$$

Torque on segment dl

$$= \vec{\mathbf{r}} \times \vec{\mathbf{F}}$$
$$= b\hat{r} \times \lambda\, dl\, E\hat{\phi}$$
$$= b\lambda\, dl\, E\hat{z}$$

Total torque on the wheel

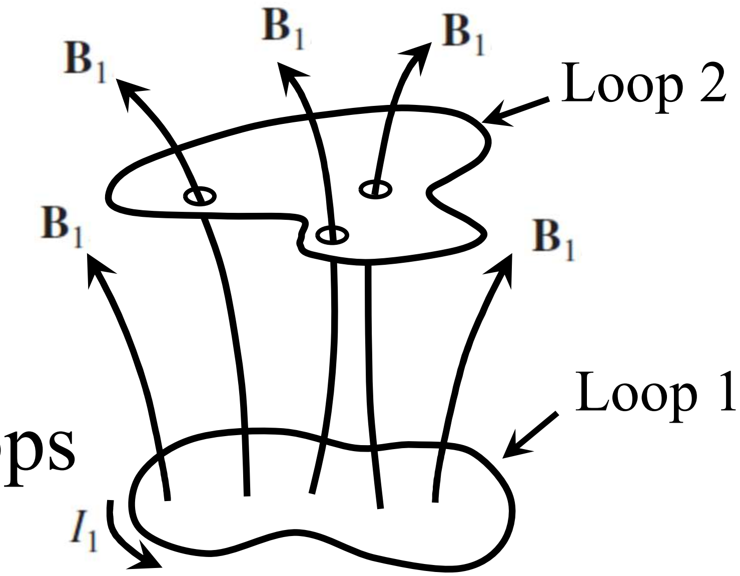
$$\vec{\tau} = b\lambda\, \hat{z} \oint E\, dl \quad = -b\lambda\, \pi a^2 \frac{dB}{dt} \hat{z}$$

Angular momentum imparted to the wheel

$$= \int \tau\, dt \quad = -b\lambda\pi a^2 \int_{B_0}^{0} \frac{dB}{dt} dt \quad = b\lambda\pi a^2 B_0 \qquad \underline{L = I\omega}$$

Mutual Inductance

$$\Phi_2 = \int \vec{\mathbf{B}}_1 \cdot d\vec{\mathbf{a}}_2 \qquad \vec{\mathbf{B}}_1 = \frac{\mu_0 I_1}{4\pi}\int \frac{d\vec{\mathbf{l}}_1 \times \hat{\boldsymbol{\imath}}}{\boldsymbol{\imath}^2}$$



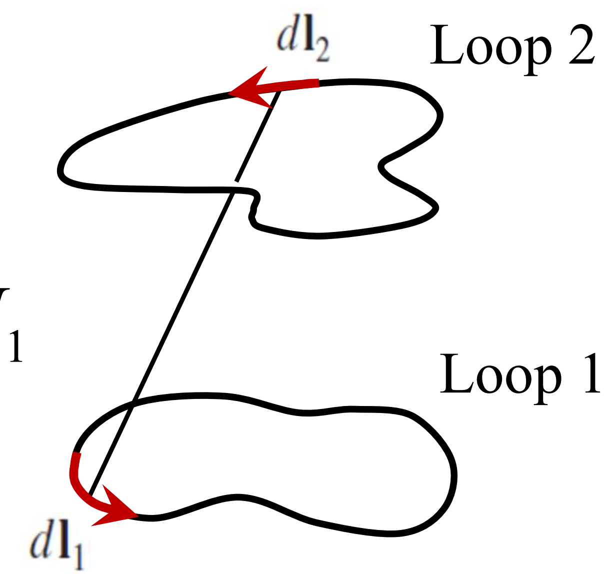
$$\mathbf{B}_1 \propto I_1 \Rightarrow \Phi_2 \propto I_1 \Rightarrow \Phi_2 = M_{21} I_1$$

M_{21}: **Mutual inductance** of the two loops

$$\Phi_2 = \int \vec{\mathbf{B}}_1 \cdot d\vec{\mathbf{a}}_2 = \int (\vec{\nabla} \times \vec{\mathbf{A}}_1) \cdot d\vec{\mathbf{a}}_2$$

$$= \oint \vec{\mathbf{A}}_1 \cdot d\vec{\mathbf{l}}_2 = \oint \left[\oint \frac{\mu_0 I_1}{4\pi} \frac{d\vec{\mathbf{l}}_1}{r} \right] \cdot d\vec{\mathbf{l}}_2$$

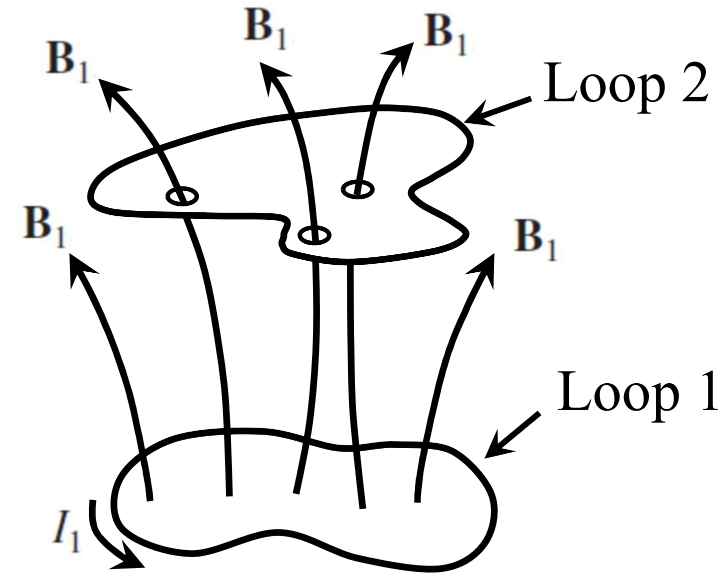
$$= \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\vec{\mathbf{l}}_1}{r} \right) \cdot d\vec{\mathbf{l}}_2 = \frac{\mu_0}{4\pi} \left[\oint \oint \frac{d\vec{\mathbf{l}}_1 \cdot d\vec{\mathbf{l}}_2}{r} \right] I_1$$



$$\Rightarrow \boxed{M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\vec{\mathbf{l}}_1 \cdot d\vec{\mathbf{l}}_2}{r}}$$

Mutual Inductance

$$\boxed{M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\vec{\mathbf{l}}_1 \cdot d\vec{\mathbf{l}}_2}{r}}$$



☞ Two imp points:

- M is a signature of purely geometrical features, size, shape, relative position etc. of the two loops
- $\boxed{M_{21} = M_{12} = M}$

☞ Let's vary current in loop 1. Flux through loop 2 will vary accordingly. Changing flux will induce an emf as per the Faraday's law:

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M\frac{dI_1}{dt}$$

Self-Inductance

☞ Changing current in a loop results in the variation of flux through itself as well.

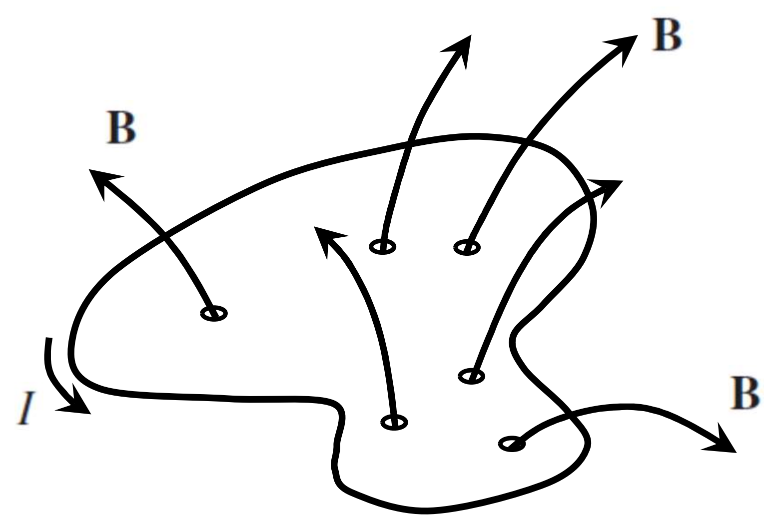
$$\Phi \propto I \quad \Rightarrow \quad \Phi = L\,I$$

L: Self-inductance (Inductance)

☞ L is a signature of the geometrical features (size and shape) of the loop.

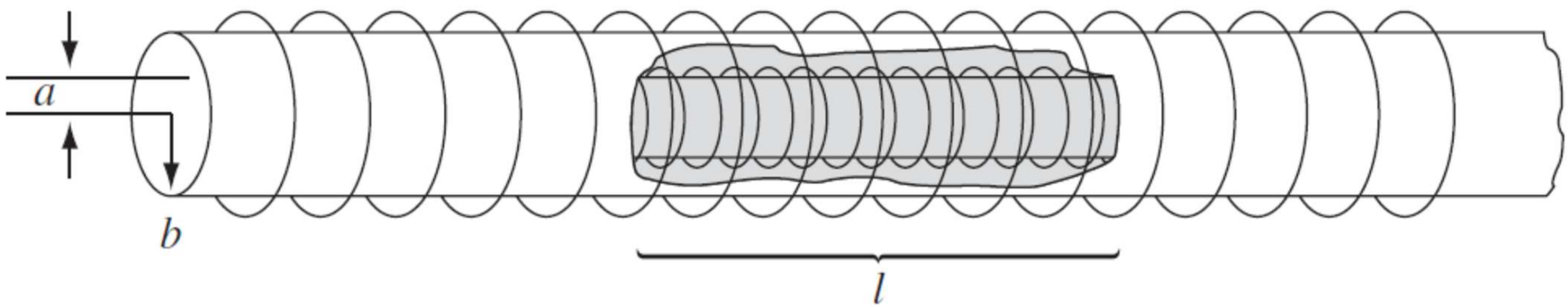
☞ Induced emf in the loop:

$$\mathcal{E} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt}$$



Mutual Inductance

Example: A short solenoid (length l and radius a, with n_1 turns per unit length) lies on the axis of a very long solenoid (radius b, n_2 turns per unit length). Current I flows in the short solenoid. What is the flux through the long solenoid?



Let us reverse the situation! Send current I through the coil of outer solenoid.

Field inside the outer solenoid: $B = \mu_0 n_2 I$

Flux through the single loop of inner solenoid: $\Phi = \pi a^2 B = \mu_0 n_2 I \pi a^2$

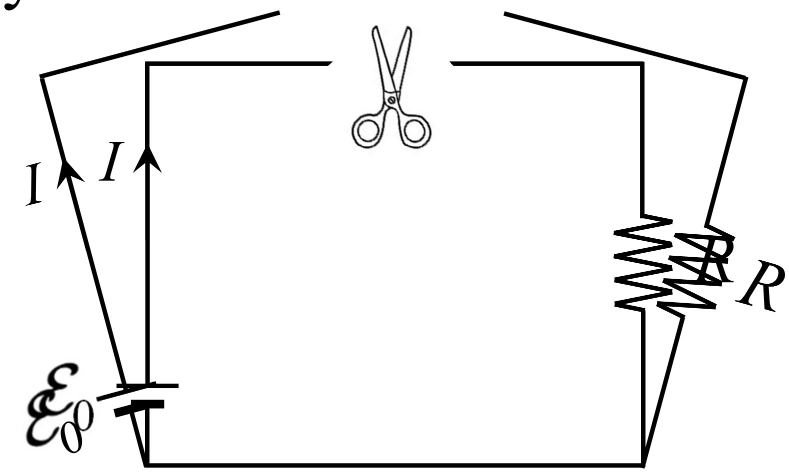
Total flux (inner solenoid has $n_1 l$ turns): $\Phi_t = \mu_0 n_2 I \pi a^2 n_1 l$

$$= (\mu_0 \pi a^2 n_1 n_2 l) I$$

$$\Rightarrow M = \mu_0 \pi a^2 n_1 n_2 l$$

Electric circuit without Inductance

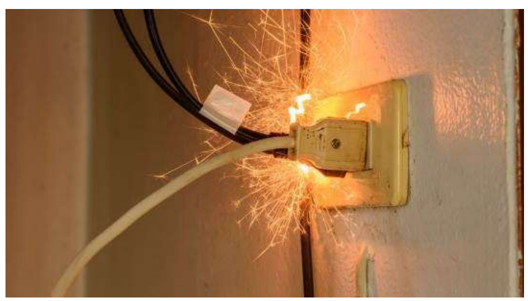
Example: Circuit carrying current is suddenly cut!



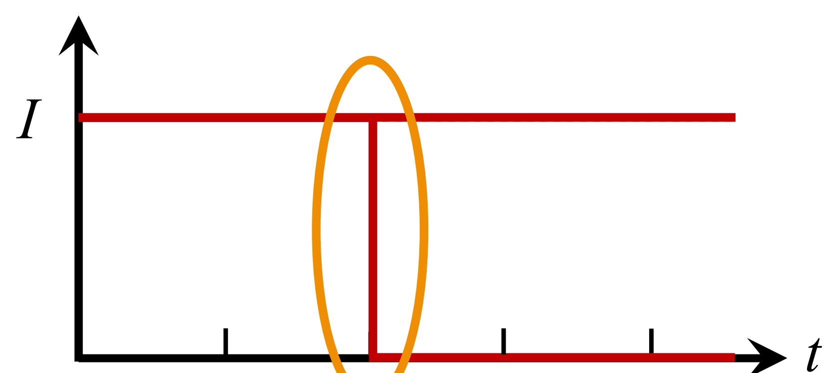
- I drops instantaneously to zero.

- At this instant dI/dt is enormous

  → Generates whopping induced EMF!

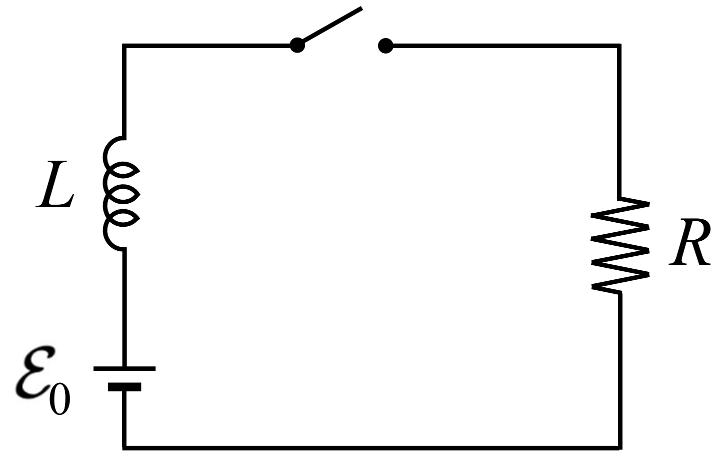


Leads to **spark** when electrical device is unplugged



Electric circuit with Inductance

Example: Find the behaviour of current flow in the circuit.



$$\mathcal{E}_0 - L\frac{dI}{dt} - IR = 0$$

$$\Rightarrow \quad \frac{dI}{dt} + \frac{R}{L}I - \frac{\mathcal{E}_0}{L} = 0$$

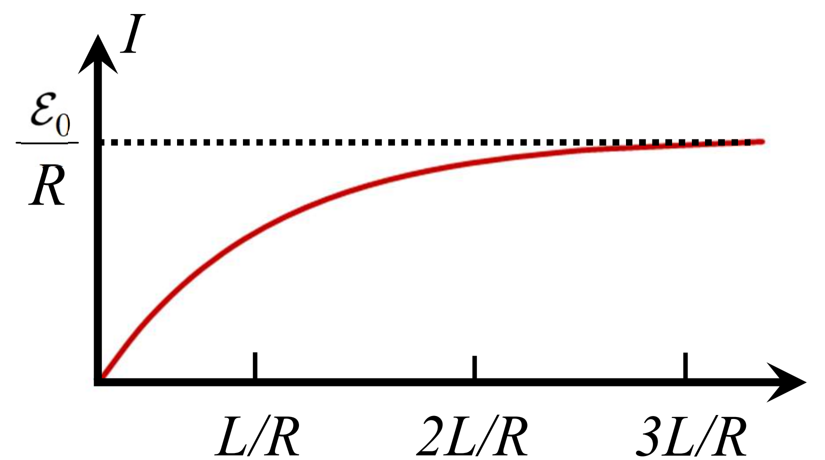
$$\Rightarrow \quad I(t) = \frac{\mathcal{E}_0}{R} + ke^{-(R/L)t}$$

Initial condition: $I(t=0)=0$ $\Rightarrow$ $0 = \dfrac{\mathcal{E}_0}{R} + k$ $\Rightarrow$ $k = -\dfrac{\mathcal{E}_0}{R}$

$$\Rightarrow \quad I(t) = \frac{\mathcal{E}_0}{R}\left(1 - e^{-(R/L)t}\right)$$ Asymptotic increase

☞ $\tau \equiv L/R$ (time constant)

☞ $L = 0$ $\Rightarrow$ I immediately jumps to $\mathcal{E}_0/R$

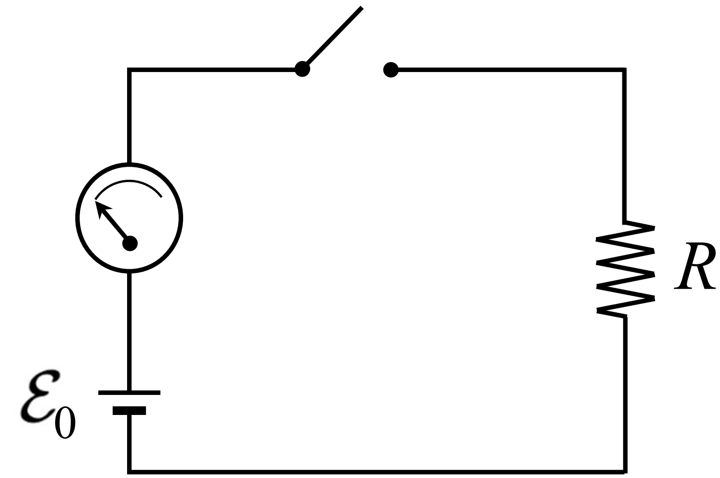


Magnetic Energy

$$dq = Idt$$

$$dW_{ext} = \mathcal{E}_{ext}dq = -\mathcal{E}_{ind}dq$$

$$= -\mathcal{E}_{ind}Idt = \frac{d\Phi}{dt}Idt$$



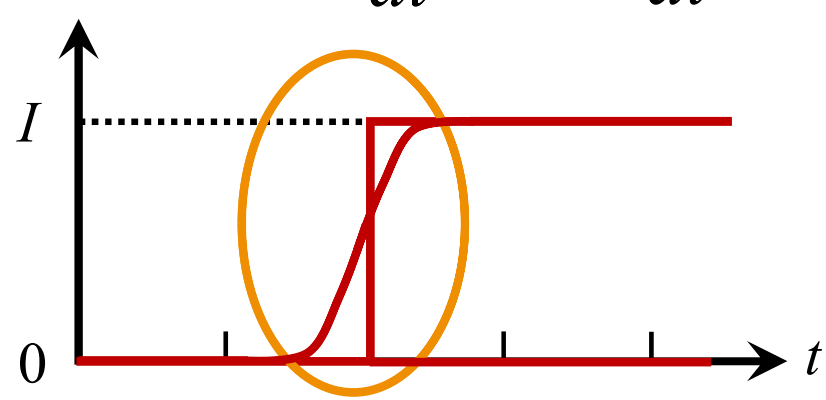
$$W_{ext} = \int dW_{ext} = \int_0^I LIdI = \frac{1}{2}LI^2 = \frac{1}{2}I\oint \vec{\mathbf{A}}\cdot \mathbf{d}\vec{\mathbf{l}}$$

$$d\Phi = L\,dI \Rightarrow \Phi = \int d\Phi = L\int dI$$

$$\mathcal{E}_{ind} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt}$$

$$\Phi = LI = \int_S \vec{\mathbf{B}}\cdot \mathbf{d}\vec{\mathbf{a}}$$

$$= \int_S (\vec{\nabla}\times\vec{\mathbf{A}})\cdot \mathbf{d}\vec{\mathbf{a}} = \oint \vec{\mathbf{A}}\cdot \mathbf{d}\vec{\mathbf{l}}$$



Magnetic Energy

$$W_{ext} = \frac{1}{2} I \oint \vec{\mathbf{A}} \cdot d\vec{\mathbf{l}} = \frac{1}{2} \oint (\vec{\mathbf{A}} \cdot \vec{\mathbf{I}}) dl \quad \Rightarrow \quad W_{ext} = \frac{1}{2} \int_V (\vec{\mathbf{A}} \cdot \vec{\mathbf{J}}) d\tau$$

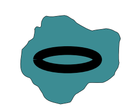
$$W_{ext} = \frac{1}{2\mu_0} \int_{all\ space} \vec{\mathbf{A}} \cdot (\vec{\nabla} \times \vec{\mathbf{B}}) d\tau \qquad \vec{\nabla} \times \vec{\mathbf{B}} = \mu_0 \vec{\mathbf{J}}$$

$$\vec{\nabla} \cdot (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = \vec{\mathbf{B}} \cdot (\vec{\nabla} \times \vec{\mathbf{A}}) - \vec{\mathbf{A}} \cdot (\vec{\nabla} \times \vec{\mathbf{B}})$$

$$\vec{\mathbf{A}} \cdot (\vec{\nabla} \times \vec{\mathbf{B}}) = \vec{\mathbf{B}} \cdot (\vec{\nabla} \times \vec{\mathbf{A}}) - \vec{\nabla} \cdot (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = \vec{\mathbf{B}} \cdot \vec{\mathbf{B}} - \vec{\nabla} \cdot (\vec{\mathbf{A}} \times \vec{\mathbf{B}})$$

$$\boxed{W_{ext} = \frac{1}{2\mu_0} \int_{all\ space} \mathbf{B}^2 d\tau} - \cancel{\frac{1}{2\mu_0} \oint_S (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) \cdot d\vec{\mathbf{a}}}$$

$$\left|\vec{\mathbf{A}}\right| \sim \frac{1}{R^2} \quad \left|\vec{\mathbf{B}}\right| \sim \frac{1}{R^3} \quad \vec{\mathbf{A}} \times \vec{\mathbf{B}} \sim \frac{1}{R^5}$$



$$\oint_S (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) \cdot d\vec{\mathbf{a}} \sim \frac{1}{R^5} \cdot R^2 \sim \frac{1}{R^3} \xrightarrow[R \to \infty]{} 0$$

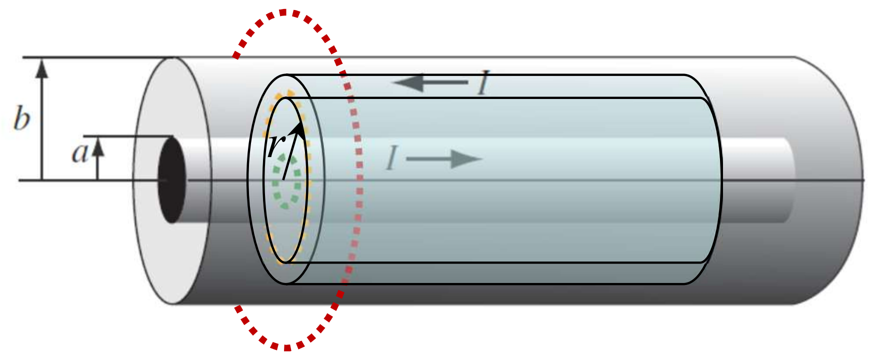
Magnetic Energy

$$\boxed{W_{mag} = \frac{1}{2\mu_0} \int_{all\ space} \mathbf{B}^2 d\tau} \qquad \boxed{W_{elect} = \frac{\varepsilon_0}{2} \int_{all\ space} \mathbf{E}^2 d\tau}$$

☞ Energy (per unit volume) stored in the magnetic field $= \dfrac{\mathbf{B}^2}{2\mu_0}$

Magnetic Energy: Example

A long coaxial cable carries current I (flows down the surface of the inner cylinder, radius a, and back along the outer cylinder, radius b). Find the magnetic energy stored in a section of length l and the self-inductance of the cable.



$$\vec{\mathbf{B}} = \begin{cases} \dfrac{\mu_0 I}{2\pi r}\hat{\phi} & a < r < b \\ 0 & r < a, r > b \end{cases}$$

Energy per unit volume: $= \dfrac{\mathbf{B}^2}{2\mu_0} = \dfrac{1}{2\mu_0}\left(\dfrac{\mu_0 I}{2\pi r}\right)^2 = \dfrac{\mu_0 I^2}{8\pi^2 r^2}$

Energy in the element: $dW = \dfrac{\mu_0 I^2}{\underset{4\pi}{\cancel{8\pi^2}}\cancel{r^2}}\cancel{2\pi}\cancel{r}\,dr\,l = \dfrac{\mu_0 I^2 l}{4\pi}\dfrac{dr}{r}$

Energy stored in the cable: $W = \int dW = \dfrac{\mu_0 I^2 l}{4\pi}\displaystyle\int_a^b \dfrac{dr}{r} = \dfrac{\mu_0 I^2 l}{4\pi}\ln\left(\dfrac{b}{a}\right)$

Magnetic Energy: Example

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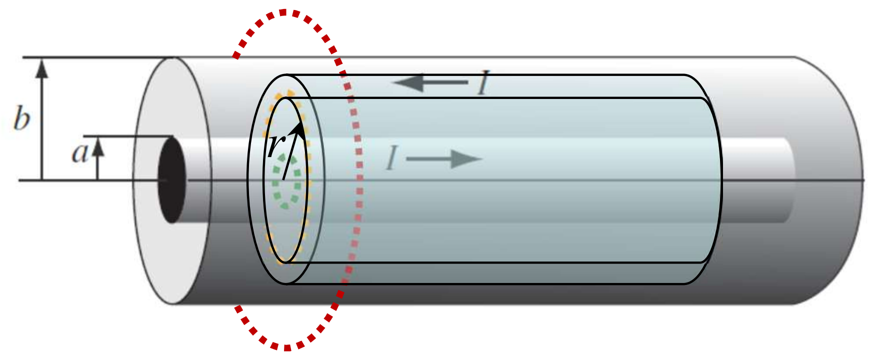

Self-inductance:

$$W = \frac{1}{2}LI^2 = \frac{\mu_0 I^2 l}{4\pi}\ln\left(\frac{b}{a}\right)$$

$$\Rightarrow \quad L = \frac{\mu_0 l}{2\pi}\ln\left(\frac{b}{a}\right)$$